T3 — Inorganic Chemistry

3.1 Periodicity — Period 3

Periodicity: The repeating pattern of physical and chemical properties of elements across periods in the periodic table. This periodicity arises because electron configurations repeat in a systematic pattern — each new period begins filling a new principal energy level.

Trends Across Period 3 (Na to Ar)

Property	Trend Na → Ar	Explanation
Atomic radius	Decreases	Increasing nuclear charge pulls electrons closer; same number of shielding shells (all in period 3)
First ionisation energy	General increase with dips at Al and S	Increasing nuclear charge. Dip at Al: 3p electron is higher in energy and easier to remove than 3s (in Mg). Dip at S: the first paired 3p electron experiences repulsion, making it easier to remove than an unpaired 3p electron (in P).
Electronegativity	Increases (Na to Cl)	Increasing nuclear charge and decreasing atomic radius means bonding electrons are more strongly attracted. Noble gases are excluded (they do not normally form bonds).
Melting point	Increases Na→Si, drops sharply at P, low P→Ar	Na, Mg, Al: metallic bonding strengthens (more delocalised electrons, smaller ionic radius). Si: giant covalent (very high mp). P, S, Cl, Ar: simple molecular structures with weak London forces (low mp).

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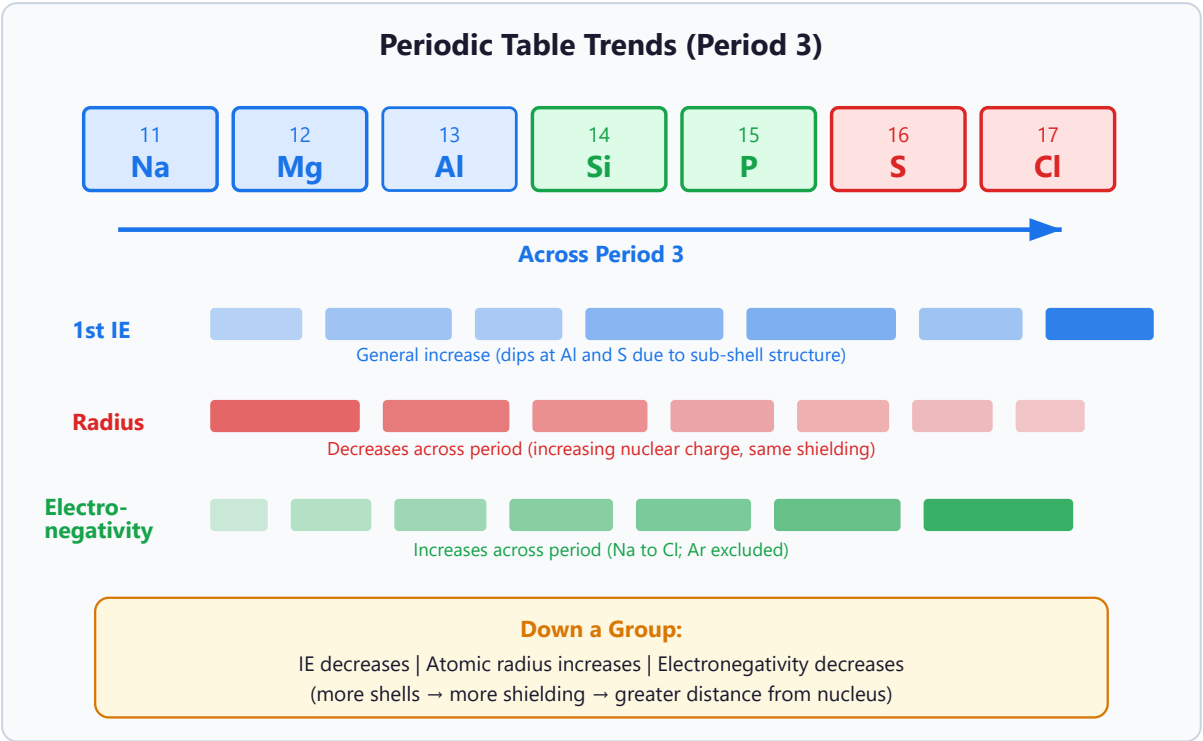
Electrical conductivity	High for Na, Mg, Al; semiconductor Si; none for P→Ar	Metals have delocalised electrons that carry charge. Si is a semiconductor (small band gap). Non-metals have no free electrons or ions.



Period 3 Oxides

There is a clear trend in the acid-base character of Period 3 oxides: metal oxides (Na_2O, MgO) are basic, Al_2O_3 is amphoteric, and non-metal oxides (SiO_2, P_4O_{10}, SO_3) are acidic.

Oxide	Structure	Bonding	Acid-Base	Reaction with Water	pH
Na_2O	Giant ionic	Ionic	Strongly basic	$Na_2O + H_2O \rightarrow 2NaOH$	~14
MgO	Giant ionic	Ionic	Basic	$MgO + H_2O \rightarrow Mg(OH)_2$ (sparingly soluble)	~9
Al_2O_3	Giant ionic/covalent	Ionic with covalent character	Amphoteric	Insoluble in water	—

SiO_2	Giant covalent	Covalent	Weakly acidic	Insoluble (reacts with hot NaOH)	—
P_4O_{10}	Simple molecular	Covalent	Strongly acidic	$P_4O_{10} + 6H_2O \rightarrow 4H_3PO_4$	~1
SO_3	Simple molecular	Covalent	Strongly acidic	$SO_3 + H_2O \rightarrow H_2SO_4$	~0

Amphoteric oxide: An oxide that reacts with both acids and bases.
$Al_2O_3 + 6HCl \rightarrow 2AlCl_3 + 3H_2O$ (with acid)
$Al_2O_3 + 2NaOH + 3H_2O \rightarrow 2Na[Al(OH)_4]$ (with base)

Period 3 Chlorides

Ionic chlorides (NaCl, $MgCl_2$) **dissolve** in water — the ions separate and become hydrated. Covalent chlorides ($AlCl_3$, $SiCl_4$, PCl_5) undergo **hydrolysis** — the covalent bonds are broken by water, producing HCl (white misty fumes in moist air) and an acidic solution.

$$SiCl_4(l) + 2H_2O(l) \rightarrow SiO_2(s) + 4HCl(g)$$

$$PCl_5(s) + 4H_2O(l) \rightarrow H_3PO_4(aq) + 5HCl(g)$$

Exam Tip: The key distinction: ionic chlorides dissolve (ions separate but no new substances form), while covalent chlorides hydrolyse (bonds break, new substances form including HCl). The white misty fumes of HCl in moist air are a key observation to mention in exam answers about covalent chloride hydrolysis.

3.2 Group 1 — Alkali Metals

Group 1 metals (Li, Na, K, Rb, Cs) have one electron in their outer s sub-shell. They are the most reactive metals, readily losing this single electron to form M^+ ions with a noble gas configuration.

Trends Down Group 1

Property	Trend	Explanation
Atomic radius	Increases	Additional electron shells
First ionisation energy	Decreases	Outer electron further from nucleus, more shielded
Reactivity	Increases	Easier to lose the outer electron (lower IE)
Melting point	Decreases	Metallic bonding weakens as ionic radius increases and charge density decreases
Density	Increases (general trend)	Mass increases faster than volume

Reactions of Group 1 Metals

With water:

$$2M(s) + 2H_2O(l) \rightarrow 2MOH(aq) + H_2(g)$$

Vigour increases down the group. Lithium fizzes gently on the surface. Sodium melts into a shiny ball, fizzes vigorously, and moves around on the surface. Potassium reacts even more vigorously and ignites the hydrogen gas with a lilac flame. Rubidium and caesium react explosively.

With oxygen: Lithium forms the normal oxide Li_2O. Sodium forms the peroxide Na_2O_2. Potassium forms the superoxide KO_2. All burn with characteristic flame colours.

With chlorine: $2M(s) + Cl_2(g) \rightarrow 2MCl(s)$. All form white ionic chlorides.

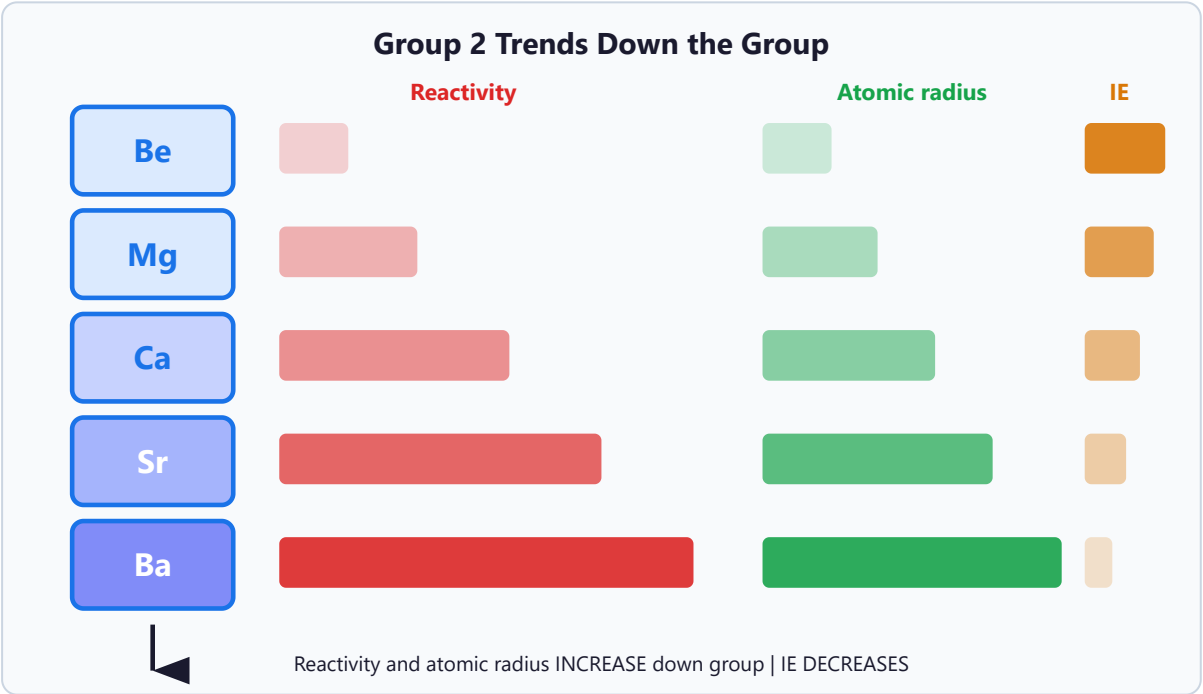
Flame colours: Li = crimson red, Na = yellow/orange, K = lilac.

3.3 Group 2 — Alkaline Earth Metals

Group 2 metals (Be, Mg, Ca, Sr, Ba) have two electrons in their outer s sub-shell. They form M^{2+} ions by losing both outer electrons.

Trends Down Group 2

Property	Trend	Explanation
Atomic radius	Increases	More electron shells
First ionisation energy	Decreases	Greater distance and more shielding
Reactivity	Increases	Easier to remove two outer electrons
Melting point	General decrease (irregular)	Weaker metallic bonding as ionic radius increases



Reactions of Group 2

With water:

$$M(s) + 2H_2O(l) \rightarrow M(OH)_2(aq) + H_2(g)$$

Magnesium reacts very slowly with cold water (forms a thin $Mg(OH)_2$ layer) but reacts vigorously with steam: $Mg + H_2O \rightarrow MgO + H_2$. Calcium fizzes steadily in cold water. Strontium and barium react increasingly vigorously.

With oxygen: $2M(s) + O_2(g) \rightarrow 2MO(s)$. All burn with characteristic flame colours: Ca = orange-red, Sr = crimson red, Ba = pale green.

Thermal Stability of Group 2 Compounds

Carbonates: $MCO_3(s) \rightarrow MO(s) + CO_2(g)$. Thermal stability increases down the group ($BaCO_3$ requires the highest temperature). This is because larger cations (e.g. Ba^{2+}) have lower charge density and are less able to polarise and distort the carbonate ion. Smaller, more highly charged cations (e.g. Mg^{2+}) have greater polarising power and can more easily destabilise the carbonate ion, making it decompose at lower temperatures.

Nitrates: $2M(NO_3)_2(s) \rightarrow 2MO(s) + 4NO_2(g) + O_2(g)$. Same trend — stability increases down the group for the same reason.

Hydroxides: $M(OH)_2(s) \rightarrow MO(s) + H_2O(g)$. Same trend applies.

Solubility Trends in Group 2

Compound Type	Solubility Trend Down Group 2	Key Examples
Hydroxides $M(OH)_2$	Increases	$Mg(OH)_2$ sparingly soluble → $Ba(OH)_2$ soluble
Sulfates MSO_4	Decreases	$MgSO_4$ soluble (Epsom salts) → $BaSO_4$ insoluble

Exam Tip: The test for sulfate ions uses $BaCl_2(aq)$ with dilute HCl first (to remove carbonates/sulfites that would give false positives). A white precipitate of $BaSO_4$ confirms sulfate is present. Do NOT use H_2SO_4 as the acid (it adds sulfate ions). The precipitate is insoluble in excess acid, confirming it is $BaSO_4$ and not $BaCO_3$.

3.4 Group 7 — Halogens

The halogens (F, Cl, Br, I) are non-metals that exist as diatomic molecules (X_2). They have 7 outer electrons and gain one electron to form X^- halide ions, or share one electron in covalent bonds.

Trends Down Group 7

Property	Trend	Explanation
Atomic radius	Increases	More electron shells
Electronegativity	Decreases	Greater atomic radius and more shielding
Boiling point	Increases (F_2 gas, Cl_2 gas, Br_2 liquid, I_2 solid)	More electrons per molecule → stronger London dispersion forces
Colour	Darkens: pale yellow → yellow-green → red-brown → grey-black (purple vapour)	—
Reactivity (as oxidising agents)	Decreases	The incoming electron is further from the nucleus and more shielded, making it harder to attract and gain an electron
Reducing power of halide ions (X^-)	Increases (I^- is the strongest reducer)	Larger ions lose electrons more easily (weaker hold on outer electrons)

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$_2$ Pale yellow gas Cl_2 Green gas Br_2 Red-brown liquid I_2 Grey-black solid Boiling Point -188°C -34°C 59°C 184°C Oxidising power Strongest oxidiser Weakest Down Group 7: BP increases (stronger London forces) | Oxidising power decreases | Colour darkens

Halogen Displacement Reactions

A more reactive (higher) halogen can oxidise (displace) a less reactive halide ion from solution. This is because the more reactive halogen is a stronger oxidising agent — it gains electrons more readily.

Halogen added	KCl(aq)	KBr(aq)	KI(aq)
Cl_2(aq)	No reaction	Orange/brown solution (Br_2 formed)	Brown solution (I_2 formed)
Br_2(aq)	No reaction	No reaction	Brown/yellow solution (I_2 formed)
I_2(aq)	No reaction	No reaction	No reaction

Example equation: $Cl_2(aq) + 2KBr(aq) \rightarrow 2KCl(aq) + Br_2(aq)$. Ionic equation: $Cl_2(aq) + 2Br^-(aq) \rightarrow 2Cl^-(aq) + Br_2(aq)$. Chlorine oxidises bromide ions; chlorine is reduced from 0 to -1, bromine is oxidised from -1 to 0.

Reactions of Halide Ions with Concentrated H_2SO_4

This reaction reveals the increasing reducing power of halide ions down the group:

- **NaCl + H_2SO_4:** Steamy white fumes of HCl. Cl^- is not a strong enough reducing agent to reduce the sulfuric acid. This is a straightforward acid-base reaction.
- **NaBr + H_2SO_4:** Initially produces HBr (steamy fumes), but Br^- can reduce H_2SO_4 to SO_2 (choking gas). Orange/brown vapour of Br_2 is also seen.
- **NaI + H_2SO_4:** Produces HI initially, but I^- is such a strong reducing agent that it reduces H_2SO_4 further to SO_2, then to S (yellow solid), and even to H_2S (rotten egg smell). Purple vapour of I_2 is also produced.

Exam Tip: The reactions of halide ions with conc. H_2SO_4 are commonly examined. Know all the observations for each halide: Cl^- gives only white fumes (HCl); Br^- gives white fumes + orange vapour + choking SO_2; I^- gives all of these plus yellow solid (S), black solid/purple vapour (I_2), and rotten egg gas (H_2S). State the oxidation state changes.

Disproportionation of Chlorine

Cold dilute NaOH:

$$Cl_2(aq) + 2NaOH(aq) \rightarrow NaCl(aq) + NaClO(aq) + H_2O(l)$$

Chlorine is simultaneously oxidised (0 → +1 in NaClO) and reduced (0 → -1 in NaCl). This is **disproportionation**.

Hot concentrated NaOH:

$$3Cl_2(aq) + 6NaOH(aq) \rightarrow 5NaCl(aq) + NaClO_3(aq) + 3H_2O(l)$$

Chlorine is reduced to -1 (in NaCl) and oxidised to +5 (in $NaClO_3$).

Disproportionation: A reaction in which the same element in one species is simultaneously oxidised and reduced, forming two products with different oxidation states of that element.

Tests for Halide Ions

Add dilute HNO_3 (to remove interfering carbonate and sulfite ions), then $AgNO_3(aq)$:

Halide Ion	Precipitate	Colour	Solubility in NH_3
Cl^-	AgCl	White	Soluble in dilute NH_3
Br^-	AgBr	Cream	Soluble in concentrated NH_3 only
I^-	AgI	Yellow	Insoluble in NH_3 (even concentrated)

The ammonia test distinguishes them further: AgCl dissolves easily in dilute ammonia because the Ag^+ forms a soluble diamminesilver(I) complex: $AgCl + 2NH_3 \rightarrow [Ag(NH_3)_2]^+ + Cl^-$. AgBr requires concentrated NH_3. AgI is too insoluble even for concentrated NH_3 to dissolve it.

Exam Tip: Always add dilute nitric acid before silver nitrate. If you use HCl, the Cl^- from HCl gives a false positive. Never use HCl or H_2SO_4 for this test. If asked how to distinguish between the three precipitates, use the ammonia solubility test.

3.5 Transition Metals

Transition metal: A d-block element that forms at least one stable ion with an incomplete (partially filled) d sub-shell. This excludes scandium (Sc^{3+}: [Ar] $3d^0$) and zinc (Zn^{2+}: [Ar] $3d^{10}$) because their ions have empty or full d sub-shells.

Electron Configurations

Transition metals fill the 3d sub-shell after the 4s. When forming ions, electrons are removed from 4s **before** 3d. Anomalous configurations: Cr = [Ar] $3d^5$ $4s^1$ (half-filled d is stable) and Cu = [Ar] $3d^{10}$ $4s^1$ (full d

is stable).

Characteristic Properties of Transition Metals

1. Variable Oxidation States

The 3d and 4s sub-shells are close in energy, so different numbers of electrons can be removed without a dramatic increase in ionisation energy. This allows multiple stable oxidation states.

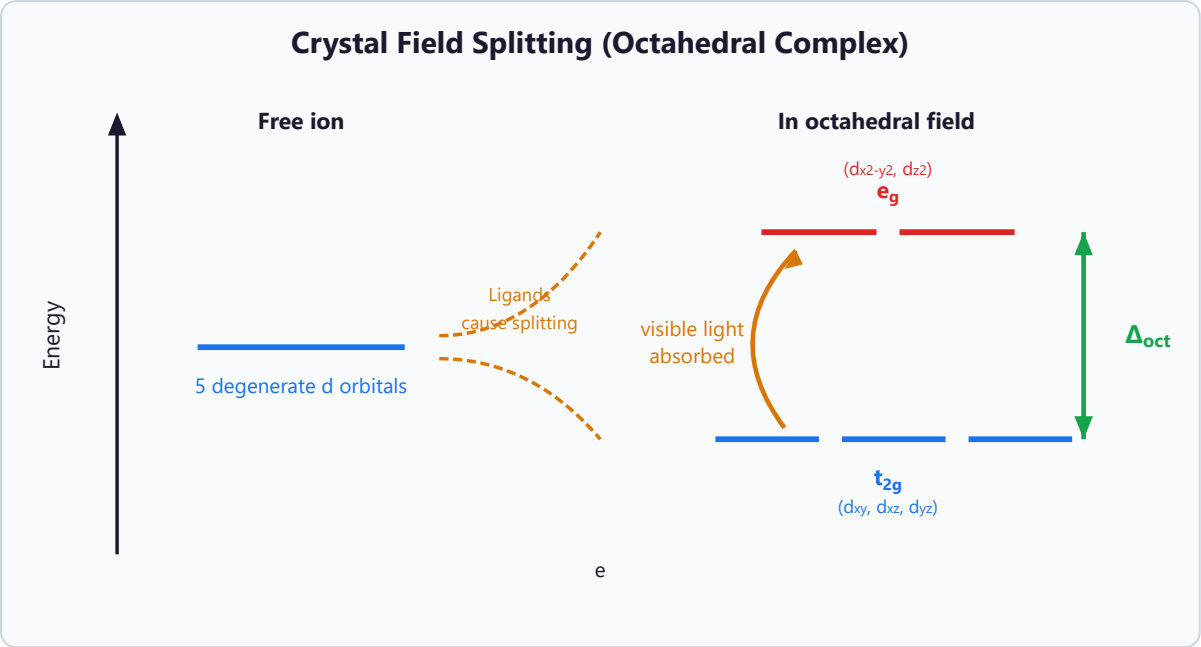
Metal	Common oxidation states	Key species
V	+2, +3, +4, +5	V^{2+} (violet), V^{3+} (green), VO^{2+} (blue), VO_2^+ (yellow)
Cr	+2, +3, +6	Cr^{3+} (green), $Cr_2O_7^{2-}$ (orange), CrO_4^{2-} (yellow)
Mn	+2, +4, +7	Mn^{2+} (pale pink), MnO_2 (black solid), MnO_4^- (purple)
Fe	+2, +3	Fe^{2+} (pale green), Fe^{3+} (yellow/brown)
Cu	+1, +2	Cu^+ (white/colourless), Cu^{2+} (blue)

2. Coloured Ions (d-d Transitions)

When ligands surround a transition metal ion, the five d orbitals split into two energy levels (crystal field splitting). Electrons absorb visible light of energy equal to the splitting energy (Δ) to transition from the lower to the upper level. The colour observed is the complementary colour of the light absorbed.

Factors affecting colour (and Δ): the identity of the metal ion, its oxidation state, the type of ligands, and the coordination number/geometry.

Why Sc^{3+} and Zn^{2+} are colourless: Sc^{3+} has $3d^0$ (no d electrons to promote), and Zn^{2+} has $3d^{10}$ (no empty d orbitals to promote into).



⁻ absorbs visible light (E = Δoct) and is promoted from t2g to eg Observed colour = complementary colour of light absorbed

3. Catalytic Activity

Transition metals are effective catalysts because their variable oxidation states allow them to accept and donate electrons, providing alternative reaction pathways with lower activation energy.

Catalyst	Process	Type
Fe	Haber process: $N_2 + 3H_2 \rightleftharpoons 2NH_3$	Heterogeneous
V_2O_5	Contact process: $2SO_2 + O_2 \rightleftharpoons 2SO_3$	Heterogeneous
MnO_2	Decomposition of H_2O_2	Heterogeneous
Ni	Hydrogenation of C=C bonds	Heterogeneous
Fe^{3+}/Fe^{2+}	$S_2O_8^{2-} + 2I^-$	Homogeneous

4. Complex Ion Formation

Complex ion: A central metal ion surrounded by ligands bonded through dative (coordinate) bonds.
Ligand: A species with a lone pair that donates it to the central metal ion.
Coordination number: The total number of dative bonds to the central metal ion.

Ligand Type	Examples	Bonds per ligand
Monodentate	H_2O, NH_3, Cl^-, CN^-, OH^-	1
Bidentate	Ethane-1,2-diamine (en), ethanedioate ($C_2O_4^{2-}$)	2
Hexadentate	$EDTA^{4-}$	6

Common shapes:

- Coordination number 6 → **Octahedral** (90°): $[Cu(H_2O)_6]^{2+}$, $[Fe(CN)_6]^{3-}$
- Coordination number 4 → **Tetrahedral** (109.5°): $[CuCl_4]^{2-}$, $[CoCl_4]^{2-}$
- Coordination number 4 → **Square planar** (90°): cisplatin $[PtCl_2(NH_3)_2]$
- Coordination number 2 → **Linear** (180°): $[Ag(NH_3)_2]^+$

Ligand Substitution Reactions

Copper(II) complexes:

$$[Cu(H_2O)_6]^{2+} + 4Cl^- \rightleftharpoons [CuCl_4]^{2-} + 6H_2O$$

Pale blue → yellow-green. Octahedral (6) → tetrahedral (4). Cl^- is larger than H_2O, so only 4 can fit.

$$[Cu(H_2O)_6]^{2+} + 4NH_3 \rightarrow [Cu(NH_3)_4(H_2O)_2]^{2+} + 4H_2O$$

Pale blue → deep blue. Coordination number stays at 6 (octahedral). Only 4 of 6 waters replaced (partial substitution) because NH_3 and H_2O are similar in size.

The Chelate Effect

Substitution by polydentate ligands (bidentate, hexadentate) is thermodynamically favourable because it increases entropy. Replacing 6 monodentate H_2O ligands with 3 bidentate en ligands increases the number of free species in solution (from 4 species on the left to 7 on the right), giving a positive ΔS and hence a more negative ΔG.

Exam Tip: When describing ligand substitution: state (1) the colour change, (2) any change in coordination number, (3) any change in shape, (4) whether substitution is partial or complete, (5) write the full balanced equation with square brackets and charges. For chelate effect questions, explain entropy increase in terms of the number of particles.

3.6 Qualitative Analysis Tests

Tests for Cations Using NaOH(aq)

Ion	Observation with NaOH(aq)	Effect of excess NaOH
Cu^{2+}	Blue precipitate $Cu(OH)_2$	Insoluble
Fe^{2+}	Green precipitate $Fe(OH)_2$	Insoluble; turns brown in air (oxidised)
Fe^{3+}	Brown precipitate $Fe(OH)_3$	Insoluble
Al^{3+}	White precipitate $Al(OH)_3$	Dissolves (amphoteric): $Al(OH)_3 + OH^- \rightarrow [Al(OH)_4]^-$
Cr^{3+}	Green precipitate $Cr(OH)_3$	Dissolves (amphoteric): $Cr(OH)_3 + 3OH^- \rightarrow [Cr(OH)_6]^{3-}$
Mn^{2+}	Pale pink/beige precipitate $Mn(OH)_2$	Insoluble; darkens in air
NH_4^+	Warm with NaOH: ammonia gas released	Gas turns damp red litmus blue

Tests for Anions

Ion	Test	Positive Result
CO_3^{2-}	Add dilute acid	Effervescence; gas turns limewater milky (CO_2)
SO_4^{2-}	Add dilute HCl then $BaCl_2$(aq)	White precipitate of $BaSO_4$

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Cl^-, Br^-, I^-	Add dilute HNO_3 then $AgNO_3$(aq)	White / cream / yellow precipitate

Flame Tests

Ion	Flame Colour
Li^+	Crimson red
Na^+	Intense yellow/orange
K^+	Lilac (view through blue glass to filter Na contamination)
Ca^{2+}	Orange-red
Sr^{2+}	Crimson red
Ba^{2+}	Pale/apple green
Cu^{2+}	Blue-green

Procedure: Clean a nichrome wire loop by dipping it in concentrated HCl and holding it in a blue Bunsen flame until no colour is produced. Then dip the wire into the sample (or into HCl and then the solid sample) and hold in the flame. Record the flame colour.

Exam Tip: Flame test colours are caused by electrons in metal ions being excited to higher energy levels by the flame energy, then falling back down and emitting light of specific wavelengths. Each metal ion has a unique set of energy levels, producing a characteristic colour. Sodium contamination (yellow flame) is very common — view potassium's lilac flame through blue glass to filter out the yellow.

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